

EXPERIMENT 12

Implement an Earley parser for context-free grammars using a simple python program.

PROGRAM:

```
import nltk

from nltk import CFG

from nltk.parse.earleychart import EarleyChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'cat' | 'dog'
V -> 'chased' | 'saw'
""")

parser = EarleyChartParser(grammar)

sentence = "the cat chased the dog".split()

print("Sentence:", " ".join(sentence))

print("\nParse Trees:\n")

for tree in parser.parse(sentence):

    print(tree)
```

OUTPUT:

```
sentence = "the cat chased the dog".split()

print("Sentence:", " ".join(sentence))
print("\nParse Trees:\n")

for tree in parser.parse(sentence):
    print(tree)
```

```
... Sentence: the cat chased the dog

Parse Trees:

(S (NP (Det the) (N cat)) (VP (V chased) (NP (Det the) (N dog)))))
```

